

```
1  /*
2  Jason Johnson
3  CS1A 1;30 - 3:50
4  11/3/25
5  LAB 7_1
6  */
7  //LAB CODE 0000
8
9  #include <iostream>
10 #include <iomanip>
11 #include <string>
12
13 using namespace std;
14
15 double getLength();
16 double getWidth();
17 double getArea(double length, double width);
18 void displayData(double length, double width, double area);
19
20
21
22 int main() {
23
24     double length{};
25     double width{};
26     double area{};
27
28     length = getLength();
29
30     width = getWidth();
31
32     area = getArea(length, width);
33
34     displayData(length, width, area);
35
36
37     return 0;
38 }
39
40
41
42
43 double getLength() {
44     double length{};
45
46     cout << "Please Enter the Length:";
47     cin >> length;
48
49     while (length <= 0) {
```

```
50     cout << "Please re-enter a value larger than 0: ";
51     cin >> length;
52 }
53 return length;
54 }
55
56
57
58
59 double getWidth() {
60     double width{};
61     cout << "Enter a value for the Width: ";
62     cin >> width;
63
64     while (width <= 0) {
65         cout << "Please re-enter a value larger than 0: ";
66         cin >> width;
67     }
68     return width;
69 }
70
71
72
73
74 double getArea(double length, double width) {
75     return length * width;
76 }
77
78
79
80 void displayData(double length, double width, double area){
81     cout << fixed << setprecision(2);
82
83     cout << endl << "Rectangle Data" << endl << "-----";
84
85     cout << endl << endl;
86
87     cout << "The Length of the rectangle is: " << length << endl;
88
89     cout << "The width of the rectangle is: " << width << endl;
90
91     cout << "The Area of the rectangle is " << area << endl;
92 }
93
94 /* OUTPUT
95 * Please Enter the Length:5
96 Enter a value for the Width: 5
97
98 Rectangle Data
```

```
99  -----
100
101 The Length of the rectangle is: 5.00
102 The width of the rectangle is: 5.00
103 The Area of the rectangle is 25.00
104
105 C:\Users\jjohnson257\source\repos\Project11\x64\Debug\
106 Project11.exe (process 11428) exited with code 0 (0x0).
107 Press any key to close this window . . .
108
109 */
110
```